
SCANNING INTO PHOTOSHOP

Unless you're using a digital camera, you're going to be scanning photographs into PhotoShop. Even if you are using a digital camera, clients will give you pictures to scan. So, you'll need a scanner, and you'll need to know how to effectively scan.

What kind of scanner do I need?

You don't need any kind of super expensive scanner for web work. Top end scanners are made for high-resolution print work. You don't want the bottom of the line junky scanner, either. Hewlett Packard makes very good scanners, the least expensive of which is about \$100.00.

Make sure that you buy a flatbed scanner, and that it comes with Optical Character Recognition (OCR) software. OCR software "reads" scanned text, placing it into an editable text file you can use for your website. OCR comes in mighty handy the first time a client hands you twenty pages of printed material and says "put this in my website".

Setting Up PhotoShop To Recognize your Scanner

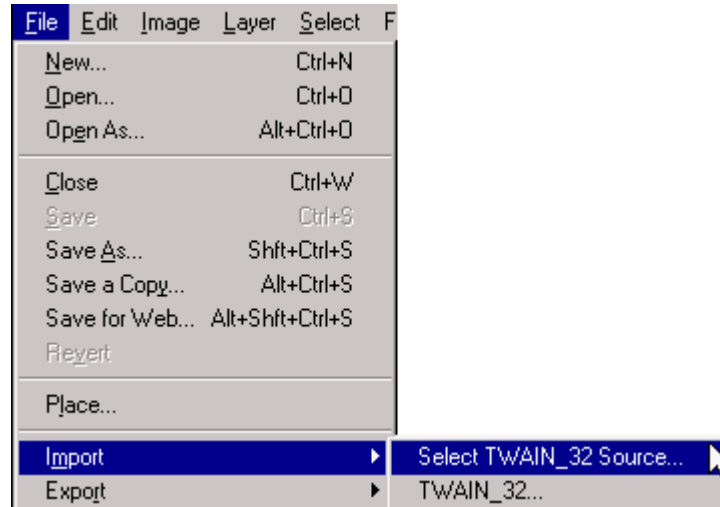
PhotoShop will not automatically recognize your scanner. Even if PhotoShop did recognize your scanner, it might not know how to talk to it. Lots of scanners are different. What you need is a program to manage the data coming from your scanner going into PhotoShop. That program is called a Data Source Manager, and happily enough, PhotoShop comes with one.

The industry standard data source manager for scanners is called TWAIN. The name "TWAIN" comes from the phrase "Ne'er the twain shall meet".

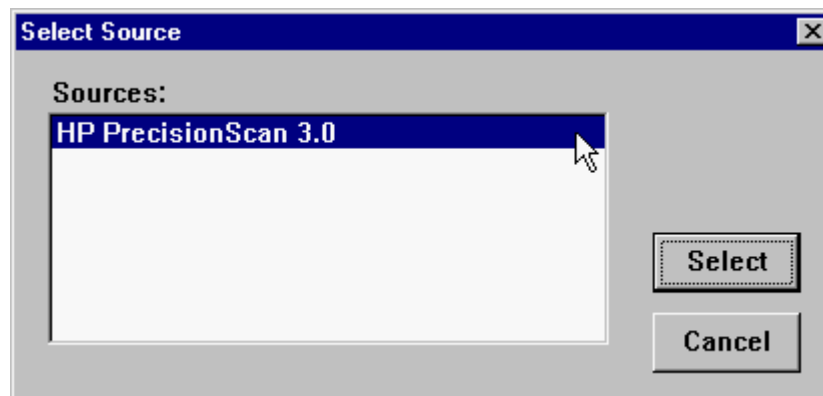
The data source manager transfers data between your scanner and your program (PhotoShop) so that they don't have to talk to each other.

Steps for Setting Up PhotoShop To Recognize your Scanner:

1. Install your scanner and all of the software that came with it.
(PhotoShop will still use your scanner's software.)
2. In PhotoShop, click **FILE-> IMPORT-> SELECT TWAIN_32 SOURCE**



PhotoShop will find all scanners you have hooked up to your computer.



3. Choose your scanner and click **SELECT**.

Now you're ready to scan into PhotoShop!

Scanning into PhotoShop

From now on, you'll scan images directly into PhotoShop.

To scan:

1. Click **FILE-> IMPORT-> TWAIN_32**

PhotoShop will find your scanner and open your scanner's scanning software. You will use the scanning software as normal, but the final image will appear in PhotoShop.

If you can't import using the TWAIN interface, use your scanner manufacturer's software to scan your images, and save the images as TIFF, PICT, or BMP files. TIFF, PICT, and BMP are all "lossless" formats, which means that they save images without a loss of data normally associated with compression. After you've scanned, then open the files in PhotoShop.

Scanner Settings

Every scanner and the software that comes with it is unique, but they all share common settings. You may have to read your manual (*ugh...*) or click Help in your scanning software if you can't find the proper setting.

Scanning Modes

Most scanners have different *modes* for scanning. The scanner will scan differently depending on what it thinks you're scanning. Many scanners today will attempt to automatically detect what kind of image you're scanning, but you may have to manually correct the scanner if it makes a mistake.

Modes:

- * **Line Art:** for black and white solid color drawings, like logos or pen drawings. Line art only reads pure black and white.
- * **Halftone:** for anything printed with tiny dots, like newspapers, magazines, or something printed on a desktop printer.
- * **Color:** For color photographs
- * **Grayscale:** For black and white photographs.

- * **Text/OCR:** For text destined for a word processor. Should not be used with PhotoShop.

Resolution

Your scanner will allow you to scan at different resolutions. When your images are published to the web, they will be **72 dpi**. You may get very good results by scanning directly at 72 dpi.

Down sampling

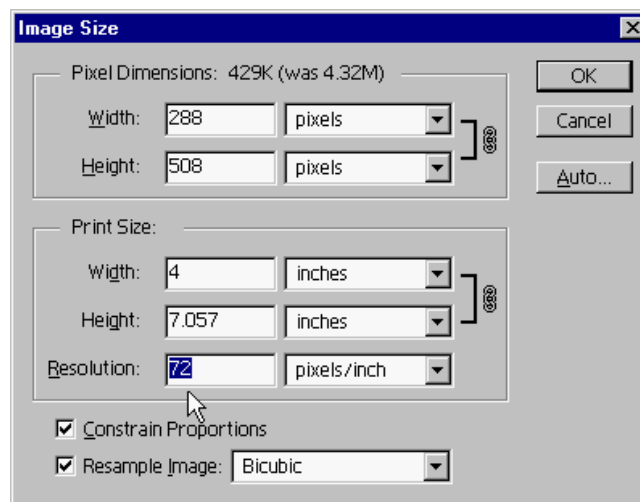
You will often get better images by scanning at a higher resolution, and using PhotoShop to *down-sample* your work to 72dpi. Curves will often look smoother, and color will often be more natural.

300 dpi is often a good starting point.

Different images and different scanners react differently to down-sampling. Scanning at 72 DPI may work just fine for you.

To down-sample an image to 72 dpi:

1. Scan image into PhotoShop at a higher resolution
2. In PhotoShop, click **IMAGE-> IMAGE SIZE**
3. Leave the check in **Resample Image**
4. Change the resolution to 72



Your final image will be the same size as your original scanned artwork.

Use Higher Resolution Images for Editing

Often, when you down sample an image, editing mistakes that you made will be hidden! When there is editing or retouching to be done, it is a good idea to do it on a higher resolution image. Then, follow the above instructions to down sample.

You'll need a pretty fast computer with lots of RAM to edit those high-resolution images, though...

Preview Mode

Most scanners have a preview mode that scans a low-resolution image first, allowing you to see and adjust what you're going to scan. Some scanners will automatically scan in preview mode if you set the resolution high.

Using Your Scanner To Enlarge Images

If you're going to enlarge an image, you're better off enlarging with your scanner instead of in PhotoShop. PhotoShop, when enlarging, adds pixels based on a mathematical algorithm. When your scanner enlarges an image, it actually scans at a higher dpi, and then spreads out the original pixels to meet your target resolution. Because all of the pixels are original, your scans will look better.

Use Enlarged Images for Editing

When you downsize an image mistakes that you made will often be hidden! When there is editing or retouching to be done, I often

Using Your Scanner to Counterfeit Money

You can use your scanner to counterfeit money by following the subsequent steps:

1. Scan money
2. Print money
3. Spend money
4. Go to jail

Image Size and Resolution

Resolution, or DPI refers to the number of dots, or pixels, on a linear inch of an image. The web displays graphics at 72 DPI, yet sometimes your client will hand you a 300 dpi image. PhotoShop can help you make the transition from a clunky, slow downloading file to speedy 72 DPI delight.

PhotoShop can also help you resize an image. Let's face it, things aren't always the way we want them. You can shrink or stretch an image to fit your liking, or you can chop out a small part of an image and make it larger.

The possibilities are endless!

Resolution

As previously mentioned, all web images should be set at 72 dots per inch. This can be achieved through scanning (scanning chapter next), or by manually changing DPI settings in PhotoShop.

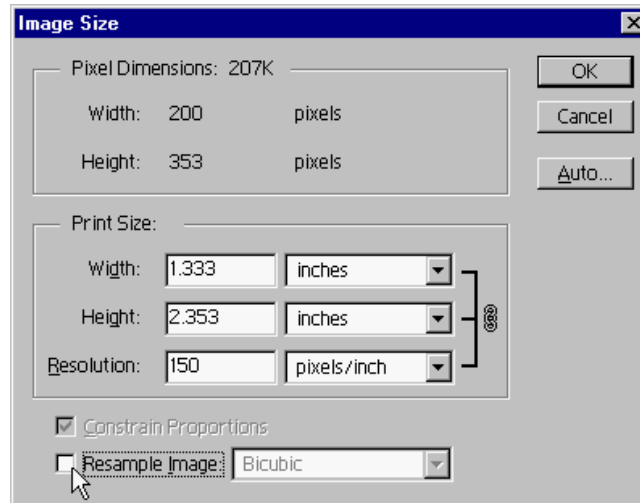
Adding or subtracting pixels is easy. Deciding what color to make the new pixels is another story. How is PhotoShop supposed to know if a new pixel is part of a line, or if it's part of someone's face? It's just a computer program. PhotoShop doesn't even know what a face is...

When changing resolution, PhotoShop runs complicated algorithms on an image to decide exactly how to change it. PhotoShop does an amazing job of adding and subtracting pixels, but it isn't magic. Usually you can only move up or down in resolution by about 1/3 before the image really suffers.

Changing Resolution (but not size) for Web Images

Sometimes you'll have an image that was scanned at a higher or lower resolution than you need. Its' size is fine, you'll only need to change the resolution.

1. Click IMAGE-> IMAGE SIZE



2. Uncheck **Resample Image**

3. Change the Resolution to 72

Image Size

When you instruct PhotoShop to change the size of an image, it merely spreads out or scrunches the image pixels. Pixels spread out—image gets bigger. Viola.

On the other hand, if you're trying to make an image larger, and still keep your current dpi, well then PhotoShop's got it's work cut out for it. To make an image larger, PhotoShop must spread pixels apart. To keep resolution the same while stretching, PhotoShop must add new pixels to the image and decide what color they should be. When shrinking an image, PhotoShop must decide which pixels to remove.

PhotoShop is really good at adding and removing pixels, but you can generally only increase or decrease size by about 1/3 before images start looking yucky.

Work on a Copy When Changing Sizes

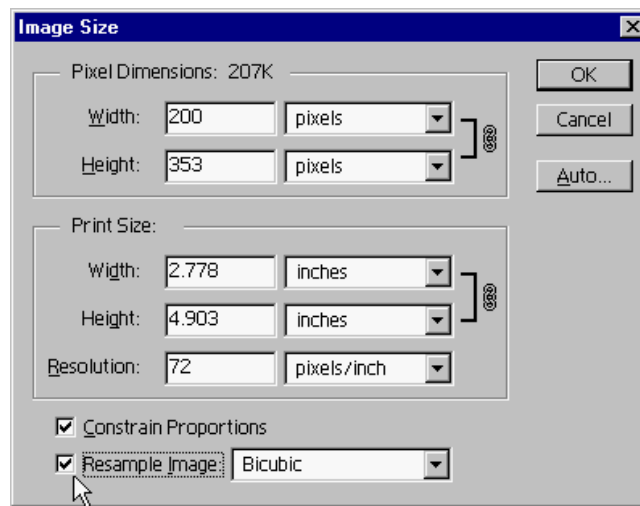
Changing image size actually changes the color information in the image, so you might want to save a copy before resizing an image forty times to decide how you like it. Every time PhotoShop changes the image, it removes or adds pixels, slightly lowering image quality.

Changing Size, but not Resolution for Web Images

1. Click IMAGE-> IMAGE SIZE

2. Make sure there is a check in **Resample Image**

"Resample Image" instructs PhotoShop to keep the resolution the same, no matter what you do to the size. If you make an image bigger, PhotoShop will have to add pixels. If you shrink the image, PhotoShop will have to remove pixels. There are three resample image choices. Use "Bicubic" it's the best.



3. Change the Pixel Dimensions of the image, not the print size.

4. Click **OK**

Heed this warning:

PhotoShop was designed for both print and web work. Thus, the program gives you the option to change print dimensions or pixel dimensions. As a web designer, you don't care about print size; your only concern is pixel Dimensions. Don't bother changing the print dimensions.

Printing Web Images

I'm sure you've tried to print something from the web, and it looked bad, didn't it?

If you positively need to print something you found on the web, you can use this trick.

Basically, you're going to shrink the image. To accomplish the downsizing, PhotoShop will move pixels closer together, resulting in a higher, more printable dpi.

1. Find the largest instance of the image you want to use for print. As you're going to have to shrink the image, the largest original will work better.
2. Click IMAGE-> IMAGE SIZE
3. Take the check *out* of Resample Image
Remember that Resample Image could have been called "Keep Resolution the Same". We don't want the resolution to stay the same; we want it to go up!
4. Change the print size to your desired print dimensions.
5. Print a copy
6. If the image quality still isn't good enough, you can bump the resolution up by following the directions for Changing Resolution (but not size) above.

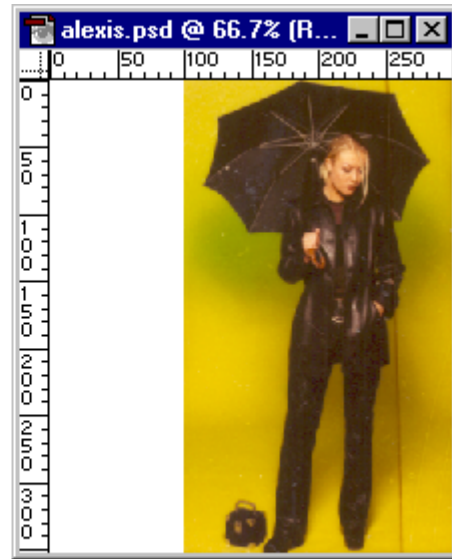
Changing Canvas Size

What if you don't want to change the image size, you just need more room in the PhotoShop window? Maybe you're trying to place two images next to each other, or add text next to an image.

Just like a painting is on a canvas, PhotoShop images are also on a canvas. If you need to add extra space in an image, you'll just change the canvas size. The picture won't get any bigger or smaller, but there will be more room around it.



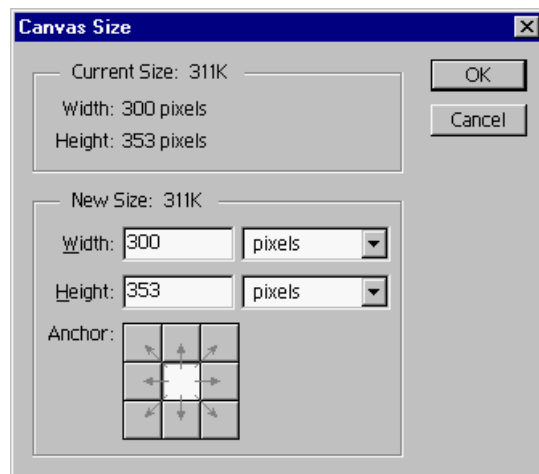
Original Image



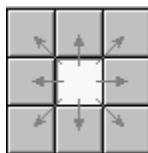
**Image with 100 pixels
added to the canvas (left side)**

To change the canvas size:

1. IMAGE-> CANVAS SIZE



2. Enter in the new Height and Width.
3. Chooses which direction to add the canvas by adjusting the anchor.



It can be tricky to add canvas in the right direction.

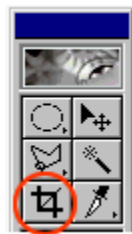
Here's the trick to keep in mind:

The arrows, *not* the white box, determine the placement of added canvas.

Cropping an Image

You've got a great image, but you only need the center. Cool, we're going to chop off the rest using the crop tool.

The crop tool is found on the toolbox.

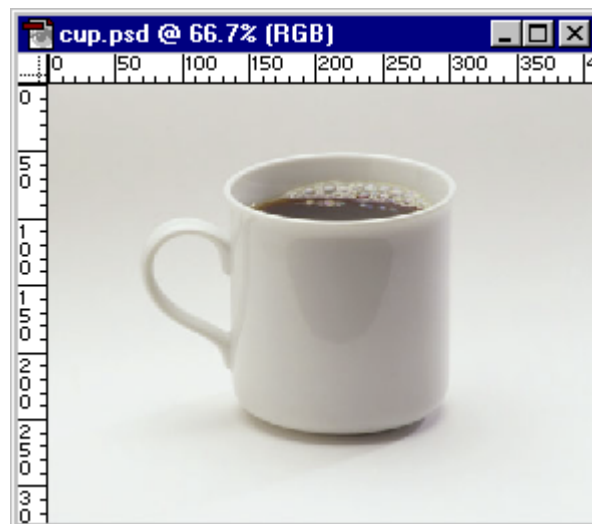


To Use the Crop Tool

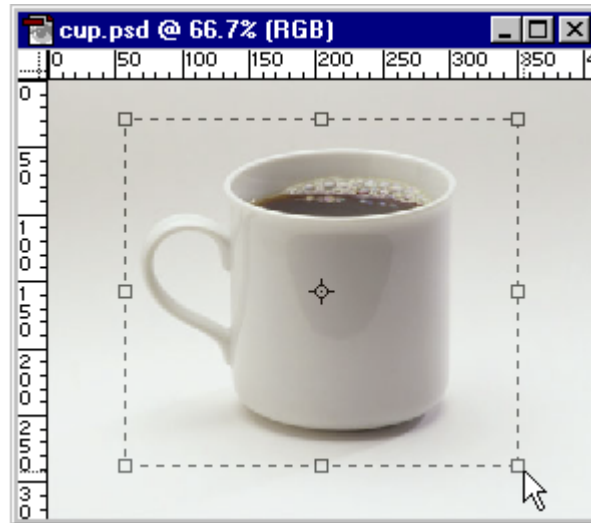
1. Open an image. I'm using **Cup.psd**.

If this were a product shot, you sure wouldn't need all that empty space on the edges.

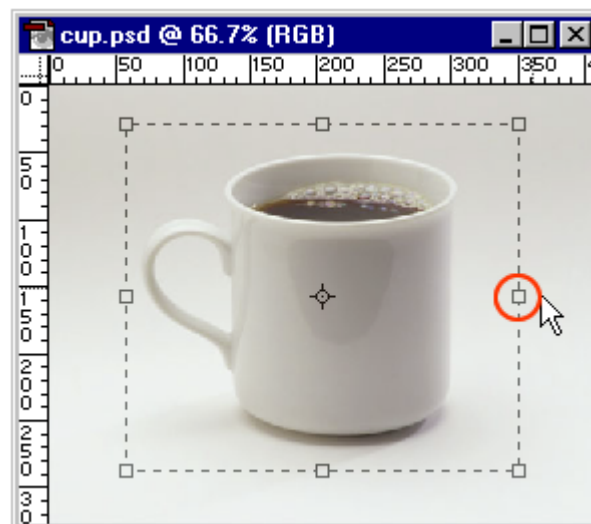
Let's crop it!



2. Select the Crop tool.
3. Drag diagonally across the cup.



4. Adjust the crop by dragging the handles (boxes) in the corners

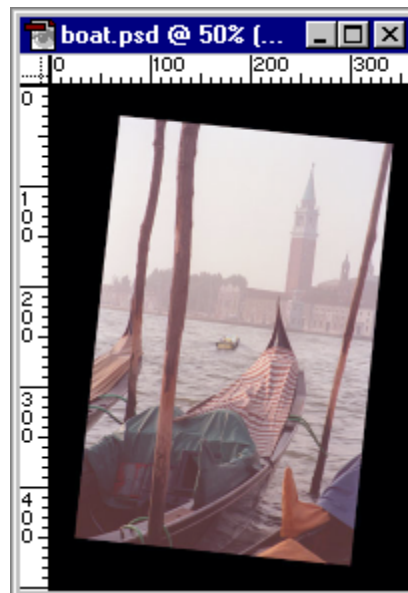


5. Hit ENTER to accept the crop
(To reject a crop and start over, hit ESC on your keyboard instead of enter.)



Using The Crop Tool To Straighten A Scan

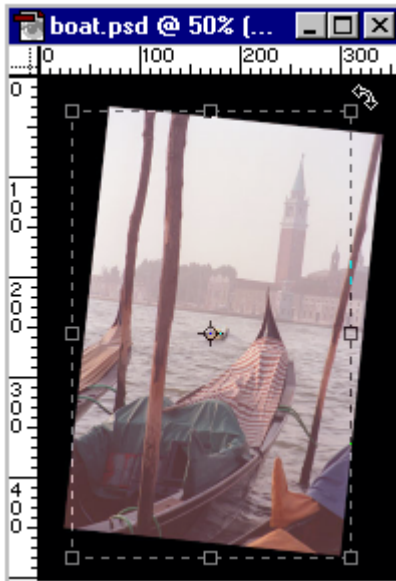
No matter how hard you try to keep an image straight in the scanner, evil gremlins will knock it off whack right before you close the scanner lid. The scan will come out slightly twisted—almost always happens.



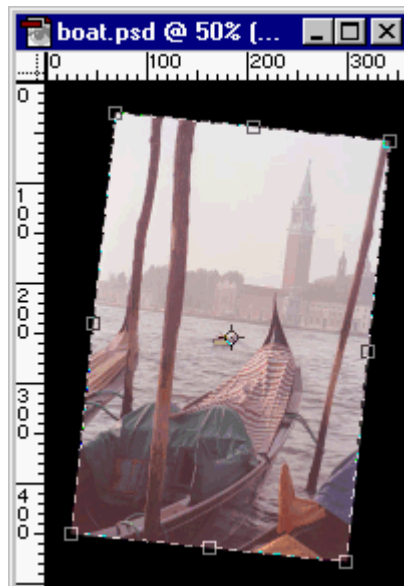
Luckily, you can foil the evil gremlins by using the crop tool.

To straighten an image:

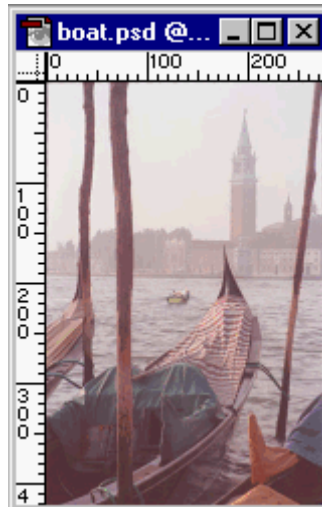
1. Select the image with the crop tool
2. Place your cursor directly above the corner handle (box). Your cursor will turn into a double arrow.



3. Click and drag in a circle to rotate the crop.
4. Adjust the crop so that the dotted lines line up with the image.



5. Hit ENTER



Viola!

Using the Crop to Enlarge Part of an Image (to Correct Proportions)

Lets say you've got a 100 pixel photograph for your website, it's a nice photo, but you don't need the outside edges... You want the image to still be 100 pixels, though. Never fear, the crop tool has anticipated your needs and has a special feature just for you. You can choose an area to crop. You'll tell PhotoShop how large that area should be *after* you crop it. PhotoShop will crop the image, and shrink or enlarge the cropped area to fit dimensions you specify.



To crop to specific dimensions:

1. Select the crop tool
2. On the Options Bar, set a target height, width, and resolution.



You can also specify that your cropped image should have the same properties as your

current image by clicking the **Front Image** button.

3. Draw your crop and hit ENTER. (or esc to cancel)

Transform Effects

Don't you want to warp, twist, resize, and stretch parts of your image? Sure, we all do.

This part of PhotoShop is really fun.

You're going to learn how to resize a part of an image to meet your needs. You'll be able to resize part of an image after you put it into another image, to make it look natural. You're going to learn how to apply perspective to a flat image. You're going to learn how to make yourself look taller, thinner, or more muscular.

Ready?

To Transform an Object

To change, or transform an object, you must first select it.

1. Open **guitar.psd**

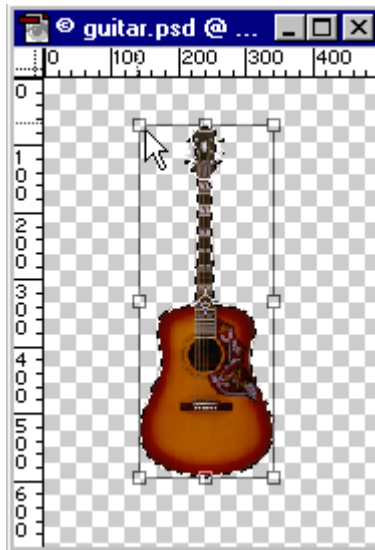
The checkerboard areas won't print. PhotoShop is just trying to tell you "there's nothing here".



2. Select the Guitar.

The easiest way to do this would be to select outside of the guitar with the magic wand and inverse the selection (SELECT-> INVERSE)

3. Click **EDIT-> TRANSFORM-> SCALE**



4. Drag one of the handles (boxes on the corners and centers of the line) to resize.
*Hold **SHIFT** on your keyboard to resize proportionally.*
5. Hit ENTER on your keyboard to accept the changes, or ESC to reject them

Transform Tools

There are a number of other tools kept under EDIT-> TRANSFORM

Rotate: Allows you to rotate an object
Adjust the rotate center point, if desired



Skew: Allows you to lean or bend an object in different ways. Useful for mapping something to a flat surface, or giving an item perspective.

Distort: Allows you to bend and push the item any way you want to. Go crazy.

Perspective: Gives an image that “it’s coming closer” or that “it’s going away” look.

Numeric: Allows you to set precise transform numbers, useful when you need something exactly this or that size, or rotated exactly a certain amount.

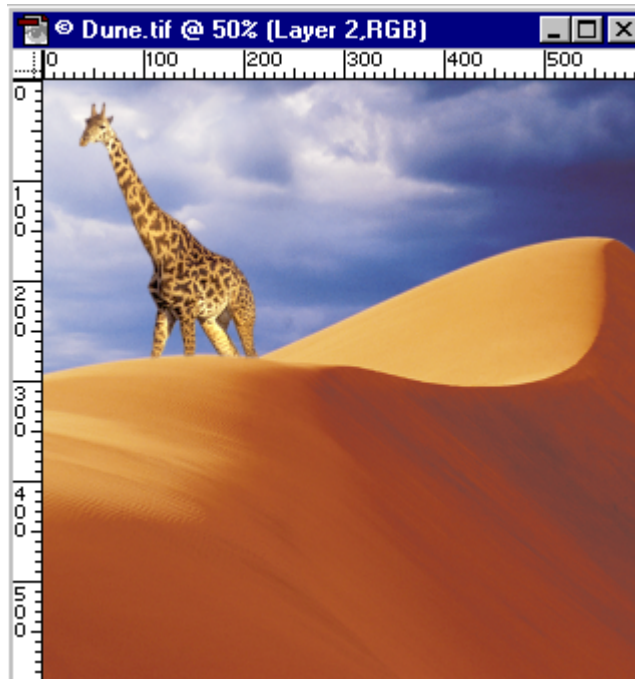
Rotate 180, 90 CW, 90CCW: allows you to instantly rotate an image 180 degrees, 90 degrees clockwise, or counter clockwise.

Flip Horizontal/Vertical: Flips an image.

Try this:

1. Open **giraffe.psd** and **dune.psd**
2. Select the giraffe and paste him into dune.
3. Click **EDIT-> TRANSFORM-> SCALE** to resize the giraffe so it looks right (if you paste an item you can immediately transform it without reselecting.)

See tips below for this exercise



Tips:

- * Cutting the giraffe out will be the hardest part of this exercise. Have patience. Use the polygonal lasso first and clean up the selection with the normal lasso.
- * Before you cut and paste contract your selection and feather by one pixel. This will help your selection look more natural when you paste. (SELECT-> MODIFY-> CONTRACT: 1 pixel) & (SELECT -> FEATHER 1 PIXEL)
- * After you paste and transform the giraffe, move him with the move tool 
- * Select the bottoms of his feet and delete them to make them "disappear" behind the dune. Don't forget to use a one pixel feather

Filter Transform Effects

PhotoShop has a number of built-in effects called Filters, later we'll use many of them for image retouching. There are some filters that come in really handy for transforming an object.

Some of the filters are useful for real-world photo editing. They can be used to turn something on a 3-D axis, or to make lips look more full. Many of the filters can be used for effect, and some look pain goofy.

Distort Filters

1. Open **Emily.tif**

We're going to really mess with this poor girl. Heh, heh, heh.

2. Select her face with the lasso. Don't worry about getting it perfect.



3. Click **FILTER-> DISTORT-> PINCH**

Crank it all the way up.



Kinda scary, isn't it?

4. Undo, and try PINCH the other way.

Pinch can also expand items and can be used to make lips look fuller, if you're into that kind of thing...

5. Try some of the other DISTORT filters on different items



**Emily, with
FILTER-> DISTORT-> TWIRL
applied.**



**The same filter applied to a
selection in a different
image.**